# *Takata Airbag Recall Data Compilation and Visualization Support*

Draft Statement of Work

693JJ926RQ000286

U.S. Department of Transportation (US DOT)

National Highway Traffic Safety Administration (NHTSA)

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## SECTION C – DESCRIPTION, SPECIFICATION, WORK STATEMENT

### 1.0 Background

The National Highway Traffic Safety Administration (NHTSA) is an agency of the U.S. Department of Transportation (DOT). NHTSA's mission is to save lives, prevent injuries, and reduce traffic-related health care and other economic costs. The agency develops, promotes, and implements effective educational, engineering, and enforcement programs with the goal of ending preventable tragedies and reducing economic costs associated with vehicle use and highway travel. NHTSA's Office of Defects Investigation (ODI) is responsible for identifying vehicle safety defects. This is done by analyzing data received from numerous sources, including consumer complaints, tips and referrals, and manufacturer-submitted data such as warranty claims, death and injury reports, and manufacturer communications and technical service bulletins. ODI also administers, on average, over 1,000 safety recalls each year, which include millions of motor vehicles, child seats, tires, and other automotive equipment.

The Takata air bag inflator recalls are the largest and most complex automotive recalls in U.S. history. These recalls involve over 60 million Takata air bag inflators in tens of millions of vehicles in the United States. The recalls are due to a design defect whereby the phased-stabilized ammonium-nitrate (PSAN) propellant used in the air bag inflator degrades, causing it to burn too quickly when ignited. This defect can result in the inflator rupturing during deployment, essentially exploding and shooting sharp metal fragments (like shrapnel) at vehicle occupants. Rupturing Takata inflators have resulted in 18 deaths in the United States and hundreds of injuries, including a significant number of serious injuries involving lacerations and other serious consequences to the occupants' face, neck, and chest area.

To address the unprecedented and varied challenges presented by the Takata recalls, NHTSA established a Coordinated Remedy Program in November 2015 to accelerate and coordinate the recalls, NHTSA appointed an Independent Monitor in December 2015 to assist NHTSA in the ongoing oversight of Takata and the vehicle manufacturers' repair programs. The Independent Monitor and his team have been heavily involved in the recalls for the past five years, and accordingly have developed a unique and nuanced understanding of the multitude of issues and challenges that the recalls pose.

### 2.0 Scope

The objective of this Statement of Work (SOW) is to enter into an agreement to acquire expert consulting services for the management of a Tableau-based data-reporting and data-visualization platform specifically tailored to the Takata recalls.

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## 3.0 Purpose

This SOW supports NHTSA's mission to save lives, prevent injuries, and reduce traffic-related health care and economic costs, as the rupturing of Takata inflators has resulted in 18 deaths in the United States and hundreds of injuries, including a significant number of serious injuries involving lacerations and other serious consequences to the occupants' face, neck, and chest area.

## 4.0 Specific Tasks

### Task 1: Takata Recall Data Analysis

The contractor shall support the intake and analysis of Takata recall data by providing:

- Management of data intake and visualization output derived from vehicle manufacturer data submissions in the Takata recalls (containing recall-related information such as completion percentages, repair activity by dealer, and unrepaired air bag inflators by ZIP code).

- Quality-control processes and data analyses, with reports accessible to registered users through a web-based portal for general reference as well as use in conjunction with owner-outreach and earned-media campaigns.

- Utilization of the Tableau platform for the management of data intake and visualization output from vehicle manufacturer submissions as described above, using this platform for analysis and to facilitate segmentation analysis of vehicle owner populations (demographic and geographic), which in turn can be used to inform potential outreach strategies.

The contractor shall:

- **Deliverable 01:** Provide NHTSA with access to data from vehicle manufacturer dashboard data submissions as incorporated into contractor's data environment (SQL and Tableau) each month.

- **Deliverable 02:** Provide NHTSA with monthly quality-control reviews of vehicle manufacturer dashboard data submissions.

- **Deliverable 03:** Coordinated monthly conference calls and make available its data science experts to provide support to NHTSA throughout the duration of the contract.

### Task 2: Project Meetings

The contractor shall participate in regular on-site or virtual project meetings to discuss status specifications, work plans, schedule and related topics. A bi-monthly status meeting shall be held with the task order COR.

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The contractor shall conduct presentations and briefings directly related to the project as requested by the government.

- **Deliverable 04:** Status meetings held every other month with the COR and other NHTSA personnel.

**Task 3 – Kick-Off Meeting**

**Resources**: Program Manager

Within fifteen days (15) of award, the contractor shall meet with NHTSA program personnel via video conference to discuss the objective and performance of the contract. During this time, the contractor shall present a briefing that discusses the detailed scope of the project:

- **Deliverable 05:** Kick-Off meeting notes and point of contact information. This report shall include:
    - Points of contact on this project
    - Background/Experience of staff who will be active on this project
    - Review and agree on status meetings
    - Schedule
    - Method (face-to-face, telephone, etc.)

Any questions or other issues regarding the technical approach and staffing plan will be addressed at this time. In addition, the administration of the contract will be discussed to ensure all parties understand their contractual duties and responsibilities.

**Task 4: Project Management and Work Plan**

The contractor shall provide a Project Manager (PM) to provide overall management of the task order, and prepare meeting notes, deliverable schedule, and biweekly activity reports.
The PM role includes, but is not limited to, the following major work streams: Project Management, Communications, Stakeholder Management, and Requirements Elicitation. The PM shall maintain daily communication and coordination with the COR when communicating with system stakeholders. The Contractor shall establish a solution that complies with Federal and DOT security requirements.

- **Deliverable 06:** Within two (2) weeks of task order award, the contractor shall prepare a Work Plan, based on the contractor's proposal, task order negotiations with NHTSA, and information discussed at the kick-off meeting. The work plan shall include:

    - Description of how the contractor will record information
    - Description of how the contractor will present/communicate findings to NHTSA

The Work Plan shall be in written/electronic format and provided to the COR two (2) weeks after task order award.

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**Task 5**: **Bi-Monthly Progress Report and Teleconference**

Beginning thirty (30) days after Contract award, the Contractor shall provide twelve (12) bi-monthly project status reports to the COR of the Contract by the 15th calendar day of the month, following the month being reported. The purpose of this report is to:

- Describe the status of the technical tasks.
- Update NHTSA on costs and schedule.

- **Deliverable 07:** The bi-monthly report shall include the following information:

**General Information**:

- Contractor name
- NHTSA contract number
- Contract number
- Project title
- Period of performance
- Reporting period
- Name and phone number for the NHTSA COR
- Name and phone number for Project Manager

**Project Status**

- Accomplishments made during the reporting period;
- Preliminary or interim results, conclusions, trends or other items of information that are of interest to NHTSA;
- Plans for accomplishments in the next reporting period;
- Problems that the contractor has experienced in conducting their services; and specific action that NHTSA should undertake to alleviate a problem;
- Remaining milestones and deliverables, and their status;
- An updated risk register;
- Projected costs for the duration of the project; and
- A comparison of actual versus projected costs

- **Deliverable 08: Pull Option** - The contractor shall provide secured (FIPS 140-2) compliant, read only connection accessible to authorized NHTSA personnel for the entirety of the period of performance (PoP). The connection will access the data via:

  - Direct database connection to all relevant database objects (e.g. tables, views)
  - Database view as defined by NHTSA, or
  - Repository contain raw data file in the format of, but not limited to:
    - CSV file with a format defined by NHTSA
    - XML file with a schema defined by NHTSA

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- **Deliverable 09: Push Option** – The contractor will email a password protected data file to selected NHTSA personnel every month and when requested by NHTSA during the period of performance (PoP). The data file will be in the format of, but not limited to:

    - CSV file with a format defined by NHTSA
    - XML file with a schema defined by NHTSA

**Deliverable/Milestone Schedule:**

The following items represent Deliverables and/or Milestones to be delivered in accordance with the following schedule:

| Item Number | Section No. | Deliverables & Milestones | Due Date |
|---|---|---|---|
| 1. Takata Recall Data Analysis | 1 | Data-intake management and analysis | Bi-Monthly |
| 2. Quality control and data analyses | 1 | Data integrity reviews | Bi-Monthly |
| 4. Data intake and visualization | 1 | Communications and findings | Bi-Monthly |
| 5. Utilization of Tableau platform | 1 | Data analyses for information about vehicle owner populations | Bi-Monthly |
| 6. Project meeting | 2 | Presentations and briefings | Bi-Monthly |
| 7. Kick-off meeting | 3 | Meeting notes and POC information | 15 days after contract award |
| 8. Project management and work plan | 4 | Work plan | 15 days after contract award |
| 9. Progress report and teleconference | 5 | Status of technical tasks and update on cost and schedule | By the 15th calendar day of every other month. |
| 10. Data and documents access | 1 | Communications and findings | Bi-Monthly |
| 11. Data and documents access | 1 | Communications and findings | Bi-Monthly |

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